2 A rescue drone is flying horizontally at a constant height of 80 m above a horizontal valley floor. The drone's speed u_D is 25 m s^{-1}.

The drone releases a supply package. The package falls under gravity.

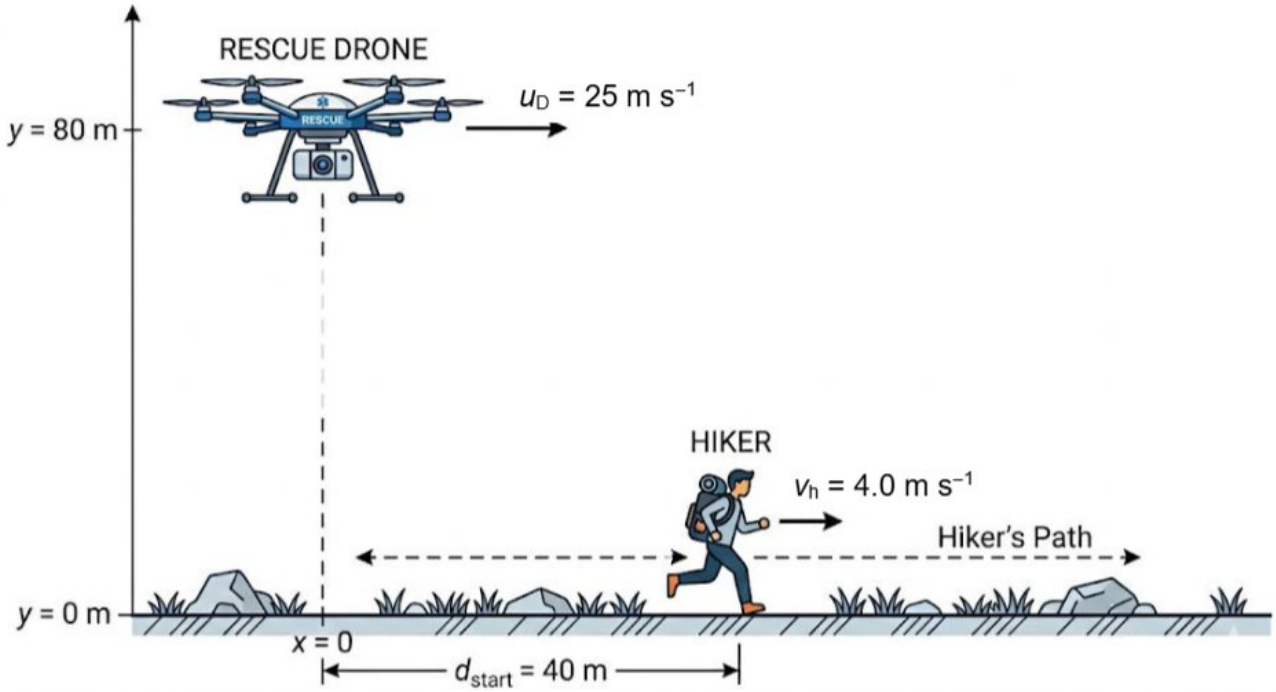


Fig. 2.1

At the instant the package is released, a hiker is horizontally in front of the drone by a distance d_{start} of 40 m (i.e., ahead along the direction of flight). The hiker immediately starts running horizontally at a velocity v_h of 4.0 m s^{-1} in the same direction as the drone's flight, attempting to catch the package.

Neglecting the effects of air resistance,

(a) calculate the time taken for the package to hit the ground,

time = ... s [2]

(b) calculate the horizontal distance the package travels from the release point to the ground,

distance = ... m [1]

(c) determine whether the hiker successfully catches the package. Show your working.

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..... [2]

[Total: 5]